

Norms as Shared Precision Priors: Values, Salience, and Cultural Transmission in Active Inference

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Abstract

Norms — shared regularities of behaviour that enable coordination among agents with heterogeneous preferences — are studied separately across game theory, cultural evolution, cognitive sociology, and active inference, with limited integration. The substrate problem is how prior preferences come to be shared across a community in the first place: how values are installed in individual agents through community-level dynamics. We address this problem within the active-inference framework. Values are identified with prior preferences (**C**) in an agent’s generative model; prestige cues and credibility-enhancing displays are formalised as precision modulators on the channel through which **C** is updated from observed behaviour; and norms are characterised as stabilised regions of high-precision shared **C** across a community. The resulting fixed-point dynamics — community priors shape salience, salience directs attention and copying, copying installs priors in next-generation agents — admit three regimes (convergence on a shared norm, bifurcation into multiple sub-community attractors, and dissolution into incoherence) determined by the precision gain of the social channel. The shape of **C** itself further governs the exploration–exploitation tradeoff in installed values. The account is complementary to existing treatments of social influence as policy-precision modulation, operating one layer up on the content of prior preferences. We propose a multi-agent simulation that operationalises the regime predictions and discuss implications for tracking how mediated information environments may restructure community-level value distributions.

1 Introduction

The cognitive and cultural substrates of norms — shared regularities of behaviour through which agents with heterogeneous preferences coordinate — have been the subject of sustained inquiry across multiple disciplines, with limited integration. Game theory and the study of conventions, descended from Lewis, frame norms as equilibria of repeated interaction [Köster et al., 2022, Oldenburg and Zhi-Xuan, 2024, Vinitzky et al., 2023]. Cultural evolution [Henrich and Gil-White, 2001, Heyes, 2018] documents the population-level mechanisms by which behaviours and values are transmitted across generations, with particular attention to prestige-biased copying and credibility-enhancing displays [Henrich, 2009, Chudek et al., 2012]. Cognitive sociology, anchored in Bourdieu’s account of habitus and developed empirically by Vaisey [2009], locates norms in durable dispositions acquired through participation in social life. Active inference [Friston et al., 2015, Parr and Friston, 2017] provides a computational vocabulary in which values are prior preferences in an agent’s generative model and attention is precision-weighting

on prediction error; multi-agent extensions of this framework [Albarracin et al., 2022, Heins et al., 2024] have begun to apply these tools to collective phenomena.

What has been missing is a formal account of the substrate that connects these traditions: the cognitive mechanism by which the prior preferences that make a norm binding for an individual agent come to be shared across a community. Equilibrium framings yield stable behaviour without specifying a cognitive substrate; cultural evolution describes the empirical transmission process in informal vocabulary; cognitive sociology characterises the phenomenology without committing to a computational mechanism; active inference has the formal apparatus but has not been systematically applied at the junction where individual generative models are coupled by social observation. The substrate problem is the gap into which this paper places its contribution.

Working in the active-inference framework, we identify values with prior preferences ($\mathbf{C}$) in an agent’s generative model. We propose that prestige cues and credibility-enhancing displays modulate the precision of the channel through which $\mathbf{C}$ is updated from observed behaviour, and that norms are best characterised as stabilised regions of high-precision shared $\mathbf{C}$ across a community. The resulting fixed-point loop — community priors shape salience, salience directs attention and copying, copying installs priors in next-generation agents — admits three regimes (convergence on a shared norm, bifurcation into echo chambers, dissolution into incoherence) determined by the precision gain of the social channel. The shape of $\mathbf{C}$ itself — sharp, broad, or what Torresan et al. [2025] call “shaped” — governs the exploration–exploitation tradeoff and the robustness of installed values.

The contribution is threefold. First, we provide a formal definition of norms as high-precision shared regions of $\mathbf{C}$, with explicit stability conditions for the loop dynamics. Second, we differentiate the account from Albarracin et al. [2025]’s treatment of social power as policy-precision modulation: we operate one layer up, on the content of prior preferences. Third, we propose a minimum viable multi-agent simulation that operationalises the regime predictions and provides a target for empirical work.

The paper proceeds as follows. Section 2 situates the substrate problem within contemporary norm research in multi-agent systems and cognitive science. Section 3 surveys the cultural-evolution backdrop, paying particular attention to the domain-generalty of prestige cues, which behave as precision parameters rather than content priors. Section 4 develops the formal account, including the definition of norms and the bifurcation conditions of the loop. Section 5 proposes a multi-agent simulation that operationalises the framework’s regime predictions. Section 6 closes with limitations and remarks on the implications for environments in which artificial agents mediate the substrate of value transmission.

2 Norm Learning Without a Substrate: The Installation Problem

Cooperative intelligence — the capacity of agents to find mutually beneficial joint courses of action — is the organising concept behind a rapidly consolidating research field. Dafoe et al. [2020] lay out the agenda: cooperation rests on understanding, communication, commitment, and norms. Of these four pillars, norms carry the heaviest load. They stabilise expectations across agents who cannot fully observe one another’s preferences, sustain coordination under incomplete contracting, and provide the decentralised scaffolding that makes institutions tractable. Whether the agents in question are AI–AI, AI–human, or human–human–via–AI, a world with more capable AI systems is a world in which norms do more work, not less [Critch and Krueger, 2020]. Getting their computational basis right is therefore not a peripheral concern for cooperative AI: it is foundational.

Norms-as-Equilibria: What Works and What Is Missing

The dominant treatment of norms in cooperative AI follows the game-theoretic tradition inaugurated by Lewis and formalised by Schelling: a norm is a self-enforcing equilibrium of behaviour sustained by mutual expectations. In this frame, a social norm *exists* when compliance and sanctioning behaviour jointly persist at population scale under coordination or social-dilemma pressure. The practical advantage of this framing is substantial. It interfaces naturally with multi-agent reinforcement learning (MARL), makes the stability conditions of norms tractable, and generates testable predictions about which equilibria persist under evolutionary pressure. The Leibo lineage of cooperative MARL papers, the decentralised sanctioning work of Vinitzky et al. [2023], and the Bayesian rule-induction approach of Oldenburg and Zhi-Xuan [2024] all inherit this framing, and all deliver real scientific progress within it.

What the equilibrium framing does not deliver is a *cognitive substrate* — a mechanistic story about what a norm is in the head of a single agent. The equilibrium frame treats the agent’s internal representation as a black box: whatever it contains, if compliance and sanctioning persist, the norm is present. Three observations mark the limits of this silence. First, violations of norms *feel surprising* in a way that violations of mere regularities do not; they elicit moral outrage, not just strategic recalculation. Second, some norms are *sticky*: agents continue to follow them even when enforcement is relaxed and even when they cannot articulate a welfare rationale. Third, and most relevant here, agents can shift from *imitating* a norm — following it to avoid sanctions — to *embodying* it, at which point the norm shapes what they attend to, what they notice as salient, and what they find imaginable as a course of action. The equilibrium account has no vocabulary for this phenomenology, because it operates at the level of policy outputs, not generative model structure.

The Köster et al. [2022] result makes this gap vivid in a particularly elegant way. Adding arbitrary “silly rules” — taboos that carry no welfare justification — into a MARL foraging environment *improves* agents’ acquisition of compliance and enforcement with welfare-relevant taboos. This result is surprising under the equilibrium frame, because silly rules add no information about the welfare-relevant equilibrium; under a precision-substrate account, it is exactly what one would expect. Silly rules supply additional Bayesian evidence that a *normative regime is in operation* — they pump up the precision of the agents’ prior over the policy distribution “follow taboos” without specifying any particular content. Once that precision is elevated, the agents learn welfare-relevant norms faster. The mechanism Köster et al. identify is not primarily about content; it is about the legibility of the normative substrate as such. This is the empirical signature of a precision-shaping story, written in MARL language.

Oldenburg and Zhi-Xuan [2024] is the most cognitively rich current treatment and the most direct dialogue partner for this paper. Their agents Bayesianly induce shared norms as discrete obligative or prohibitive rules from observed compliance and sanction data; they show that a *shared assumption* that a norm system is operating is sufficient to bootstrap common knowledge of specific norms, enabling rapid convergence and stable transmission to newcomers. This is a genuine advance: unlike equilibrium-only accounts, it gives norms propositional structure in individual agents. But the “shared assumption” itself remains unmodelled prior content. It is posited as an external parameter of the inference problem rather than as something agents acquire from a community, transmit with fidelity, and maintain through coordinated attention. Representing norms as discrete inferred rules also forecloses the graded, phenomenologically thick account of norm-internationalisation that the sticky-norm and violation-surprise phenomena require. The rules are the outputs of inference; the question of what *installs the prior* that makes the shared assumption credible is left open.

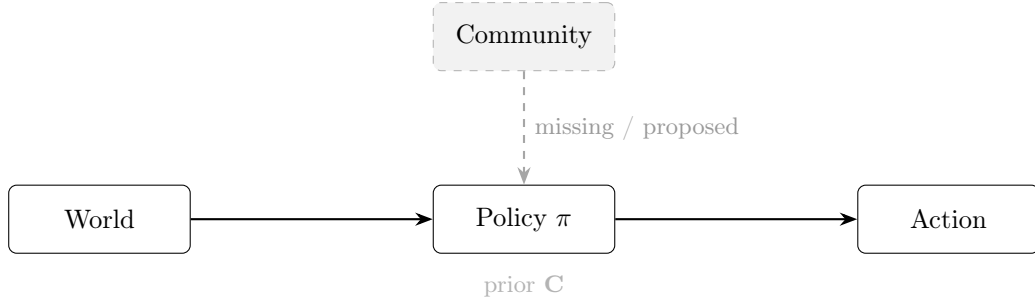


Figure 1: The missing arrow: existing cooperative-AI norm models operate on the horizontal chain (world $\rightarrow$ policy $\rightarrow$ action) and leave the community-to-prior channel exogenous (dashed). This paper’s contribution is to formalise the shaded arrow: how community-level precision dynamics install an agent’s prior preferences over outcomes ($\mathbf{C}$), constituting the cognitive substrate of norms.

The Installation Question

Where do an agent’s priors over what-counts-as-a-norm come from? This is the question the equilibrium and Bayesian-rule accounts both treat as exogenous. Norms are handed to agents, or emerge from iterated play, but in neither case does the account say how one generation’s normative landscape gets written into the next generation’s starting priors. In human cognition and cultural evolution, this is precisely the question that *is* answered: credibility-enhancing displays, prestige-biased transmission, and community-level attention allocation are the mechanisms by which prior preferences over outcomes are shaped and reproduced [Henrich, 2009]. Any formal account of norm learning that cannot speak to these mechanisms is incomplete at the level of mechanism, even if it is complete at the level of equilibrium description.

The stakes of this incompleteness become acute when AI systems participate in human normative life. Kulveit et al. [2025] argue that incremental AI integration can gradually disempower human populations not through any adversarial takeover but through structural drift: as AI systems become load-bearing participants in economic, cultural, and institutional life, the implicit alignment between human interests and system behaviour that comes from human participation being generative is progressively eroded. Under the equilibrium framing of norms, AI systems that sit at the behavioural equilibrium look unproblematic. But if norms supervene on shared cognitive architecture — if their stability is grounded in the precision-shaping dynamics through which a community installs prior preferences in its members — then AI systems that participate only at the policy level, without contributing to the substrate that makes norms credible, can hollow out the normative structure even while the behaviour looks intact. The installation question is therefore not merely a technical gap in formal models; it is the micro-level mechanism through which gradual disempowerment operates.

Differentiation from Albarracin et al. (2025)

Albarracin et al. [2025] formalise social power as the differential capacity to set the policy-precision parameter (γ) in others’ generative models. In their account, a socially powerful agent is one who can raise or lower how confidently another agent pursues a given policy — shaping the effective temperature of the policy distribution without necessarily altering what the agent considers valuable in the first place. This is a significant contribution: it gives power a formal active-inference definition, it engages Bourdieu’s concepts of habitus, field, and capital, and it shows how Foucauldian disciplinary mechanisms can be read as precision-modulation operations. Our move is one layer up, on the *content* of prior preferences ($\mathbf{C}$) — that is, what an agent comes to consider valuable in the first place, rather than how confidently it pursues a given policy. Both

layers are causally involved in the cultural reproduction of norms: a community that controls the policy-precision parameter γ in its members (the Albarracin layer) can enforce compliance, but a community that controls the content of $\mathbf{C}$ (our layer) shapes what counts as desirable before any enforcement becomes necessary. The prior-preference content layer is, we argue, the substrate that cooperative-AI norm theory currently lacks. Albarracin et al. provide the complementary half of the story; this paper provides the other half, and together they constitute a two-layer account of normativity in active-inference agents. The dynamics by which epistemic communities stabilise or fragment around shared $\mathbf{C}$ -vectors are developed in Albarracin et al. [2022], which we build on directly in Section 4.

Roadmap

The remainder of the paper develops the missing arrow in three steps. Section 3 establishes the cultural-evolution backdrop, reviewing the mechanisms — prestige-biased transmission, credibility-enhancing displays [Henrich, 2009], and community-level attention allocation — through which prior preferences are differentially shaped and reproduced across agents. Section 4 develops the formal account: we define norms as stabilised regions of high-precision shared $\mathbf{C}$, characterise the installation channel as a community-level precision-shaping process, and derive stability conditions for the resulting attractor landscape. Section 5 proposes a minimal simulation instantiating the account in a pymdp multi-agent environment, designed to exhibit the bifurcation between norm-convergence and echo-chamber fragmentation as a function of the precision dynamics.

3 Where Values Come From: Prestige, CREDs, and the Cultural-Evolution Substrate

Humans are, in a fundamental sense, obligate cultural learners: the overwhelming share of what any individual knows, wants, and treats as worth attending to was acquired from other people rather than rediscovered from scratch [Henrich and Gil-White, 2001]. This dependence runs deeper than the transmission of practical skills or factual beliefs. What counts as a *value* — which outcomes are desirable, which actions are admirable, which objects are sacred or disgusting — is itself downstream of what gets attended to, imitated, and rehearsed in a community’s daily life. A child who grows up watching elders fast, or observing neighbours donate conspicuously to shared projects, or noticing which adults attract the gaze and proximity of many others, acquires not just a collection of beliefs but a structured landscape of salience: some things matter a great deal, others barely register. The cultural-evolution tradition, crystallised in Henrich and Gil-White [2001], has spent the past quarter-century documenting the population-level mechanisms that produce and maintain these landscapes. This section surveys that account in enough detail to reveal where a formal bridge to predictive processing is both possible and necessary.

Prestige as a coordination device for collective attention

The central conceptual move in Henrich and Gil-White [2001] is the distinction between *prestige* and *dominance* as two structurally different routes to social influence. Dominance hierarchies, which appear across social primates, are sustained by intimidation: individuals defer to dominant animals to avoid costs. Prestige, by contrast, is *freely conferred deference*. Copiers voluntarily attend to, sit near, gift, and imitate high-prestige models because doing so raises their own fitness by providing access to superior information or skill. The evolutionary logic is straightforward: when acquiring complex cumulative culture requires accurately identifying good models, individuals who can cheaply infer model quality by reading others’ deference choices will outcompete those who must independently assess each candidate. The community’s existing attention

distribution therefore functions as a *second-order prior* over whom it is worth attending to — a public signal that learners consult before committing their own attentional resources.

Henrich et al. [2015] formalise this logic in a culture–gene coevolutionary model of “big man” leadership. In small-scale egalitarian societies, the big man acquires followers not through coercion but through accumulated demonstrations of generosity, oratory, and successful coordination. Critically, followers copy big-man behaviours (including prosocial ones), which in turn creates selection pressure for big men who are genuinely prosocial — because their only currency is the continued attention and copying their conduct commands. The model shows that this prestige channel produces phenotype correlation between leaders and followers and generates prosocial norms as stable downstream outcomes: the attention asymmetry is not merely a readout of pre-existing value distributions but actively reshapes them. Egalitarian levelling mechanisms — gossip, mockery, ostracism, and in extremis execution — function not as alternatives to prestige dynamics but as its complement: they *withdraw* community attention from would-be dominants, enforcing the condition under which prestige (rather than coercion) remains the operative currency. Flat societies are densely attention-managed, not attention-neutral.

The clearest experimental demonstration of how this second-order attention mechanism works at the level of individual learning is Chudek et al. [2012]. In a series of studies with 3–4-year-old children, a purely *attentional* cue — ten seconds of differential bystander gaze toward one of two adult demonstrators, with no other information about their competence or history — more than doubled the probability that children subsequently copied that demonstrator on a novel task. Crucially, the effect *generalised across domains*: prestige established through others’ gaze in one context (e.g., artifact use) transferred to unrelated copying choices (e.g., food preferences). This domain-generalty is theoretically decisive. A content-specific prior — “this person knows about artifacts” — would not transfer to food choices; the prestige cue would stay in the domain where it was established. What does transfer is a *precision signal*: “this person’s behaviour is worth attending to, period.” In predictive-processing terms (though the authors do not use this vocabulary), the bystander gaze cue behaves like a generalised precision parameter rather than a domain-restricted content prior. It raises the weight accorded to the high-prestige model’s actions across the entire space of observable behaviours, without specifying what those actions will contain. Section 4 formalises this observation precisely.

CREDs and the costliness logic

Once language evolved to allow the cheap transmission of beliefs, cultural learners faced a new inference problem: which professed values does a model *actually hold*, and which are mere verbal performance? Henrich [2009] argues that natural selection favoured learners who weighted models’ expressed beliefs by the costliness of their observable commitment to those beliefs — what he calls Credibility-Enhancing Displays (CREDs). Fasting, painful initiations, celibacy, martyrdom, and public sacrifice are actions whose costs make sense only if the agent genuinely holds the associated belief or value; they are costly signals that cannot be cheaply faked. Learners who copied the values of agents who performed CREDs systematically acquired the values of genuinely committed models, rather than those of convenient verbal performers.

The fit with precision-weighted inference is near-direct, and Section 4 develops it explicitly. Informally: without CREDs, a model’s verbal claim about a value provides low-precision evidence — it could reflect genuine commitment or could be strategic posturing. A CRED raises the precision of the channel from observed-other-behaviour to one’s own value-prior. High-CRED demonstrations are high-confidence messages about what the transmitter actually values; low-CRED environments produce noisy, uncertain updates. Empirical work by Lanman and Buhrmester [Lanman and Buhrmester, 2017] has confirmed this logic: childhood exposure to parental CREDs predicts adult theism more robustly than competing accounts based on cognitive style or existential anxiety, suggesting that what gets installed is not a bare belief but

a *calibrated prior* whose strength tracks the evidential quality of the exposure. The CRED mechanism and the prestige mechanism are thus complementary channels in the same installation system: prestige cues direct *which* models receive attention; CREDs modulate *how much confidence* to accord to those models’ expressed commitments.

Attentional infrastructure is itself culturally inherited

A natural objection at this point is that while the *contents* of attention may vary culturally, the *mechanisms* that structure attention — what kinds of agents and events are even candidates for imitation — must be evolutionarily fixed. Heyes [2018] mounts a sustained argument that this objection is wrong. Her “cognitive gadgets” thesis distinguishes a domain-general genetic “starter kit” of social attentional biases (sensitivity to faces, voices, and social rewards) from the higher-order mechanisms of selective social learning — including imitation, mind-reading, and norm compliance — which are themselves culturally assembled during development. The decisive piece of evidence for present purposes: there is empirical support for the claim that humans can *socially learn attentional biases*, specifically by learning to attend more to some agents than others through the mechanism of following third-party gaze. That is, the same bystander-gaze cue that Chudek et al. use as an experimental manipulation of prestige is also the mechanism by which children acquire durable attentional dispositions — what to find exemplary, whom to watch, which features of the social environment carry signal. The salience landscape is not merely populated through cultural transmission; its *topography* — which gradients channel attention up rather than down — is itself inherited.

Heyes [2024] extends this framework to norm psychology specifically. Rather than treating norm-following as a genetically fixed motivational module (“normative competence as instinct”), she argues that normative competence is a culturally assembled cognitive gadget built atop domain-general attentional and motivational capacities. Crucially, what counts as a norm — which regularities in others’ behaviour are picked up as prescriptive rather than merely descriptive — is not given in advance but is itself acquired through the same prestige-and-gaze mechanisms that structure learning more generally. The implications for any account of value transmission are significant: there is no innate, domain-general “norm receptor” that picks out which social patterns to treat as obligatory. Instead, the very categories through which norms are perceived and reproduced are part of what gets transmitted. This recursive structure — attentional dispositions shaping what gets copied, which in turn shapes attentional dispositions in the next generation — is precisely the loop this paper seeks to formalise.

The ceiling of the cultural-evolution account

The literature surveyed in this section provides a rich *empirical* story about value transmission: which individuals attract community attention, under what conditions, through what cues, with what consequences for the behavioural repertoires of those who copy them. The account is anchored in experimental data [Chudek et al., 2012], population-genetic formalisms [Henrich et al., 2015, Egozi and Ram, 2024], and developmental evidence [Heyes, 2018, 2024]. What this literature does *not* provide is a vocabulary that connects these dynamics to the computational architecture of the individual learner: why does bystander gaze function as a domain-general precision signal rather than a domain-specific content cue? What is happening, formally, when a CRED raises the confidence with which a value-prior is adopted? How do community-level patterns of attention crystallise into the stable configurations we call norms? These questions ask for a level of mechanistic description that cultural evolution’s largely informal vocabulary does not supply. The next section takes up precisely this gap, mapping prestige- and CRED-mediated attentional dynamics onto the formal language of precision-weighted predictive processing, and showing how the loop that closes from community priors through attention and copying back to community priors can be given a rigorous active-inference rendering.

4 Values as Saliency Priors: A Formal Account

4.1 Attention is precision; saliency is expected information gain

The first step is to refuse a conflation that even active-inference researchers routinely commit. Feldman and Friston [2010] establish the load-bearing identification: *attention* is the inference of precision Π on prediction errors — a gain on the likelihood mapping that determines which channels of current sensory evidence dominate posterior updating. Higher precision at a level of the hierarchy means a louder error signal at that level, and louder errors do more work in revising beliefs. Saliency is a different quantity. Parr and Friston [2017, 2019] draw the distinction with full formality: saliency is the *expected information gain* associated with a candidate sample of the world — prospective and policy-conditional, not retrospective and stimulus-bound. To a first approximation, attention is precision over the data the agent is *currently* consuming; saliency is a forward-looking quantity that scores, for each candidate next observation, how much the agent’s posterior would contract were that observation actually obtained.¹ Saliency is therefore not a property of the distal world. It is a property of the prior-shaped uncertainty landscape an agent carries around with it.

The framing matters because, once one allows that deep priors set the precision weights distributed across the hierarchy, saliency inherits its shape from those priors. Parvizi-Wayne [2024] makes the point most forcefully: under active inference, the endogenous–exogenous attention dichotomy collapses. There is no precision-free signal entering the cognitive system. What looks like bottom-up, stimulus-driven capture is in fact the allocation of precision against prediction errors generated by deep generative expectations the agent already holds. The corollary, due to Limanowski and Friston [2018], is that phenomenal transparency and opacity are themselves precision phenomena — when precision on a level is high and stable, the agent “sees through” it; when precision is low or unsettled, the level itself becomes available to introspection. Together, these results license the move at the heart of this paper: *deep priors* — *including the agent’s preferences* — *shape what becomes salient*. Ransom et al. [2020] press the strongest empirical objection to attention-as-precision: affect-biased attention orients the system toward emotionally significant stimuli even when their feature-level precision is low. We engage this directly because it is not in fact a counter-example. Their data are consistent with a deep-priors-set-precision rebuttal: emotionally significant stimuli are exactly the ones for which the agent has high-precision priors over their value-relevance, and those high-precision deep priors propagate down the hierarchy as gain modulation on the lower-level error signals that constitute affect-biased attention. The Ransom et al. result, in other words, is evidence *for* the framing this paper develops, provided one is willing to grant that values sit upstream of sensory precision-setting — which is exactly the move we now make.

4.2 Values as prior preferences $\mathbf{C}$

Active inference furnishes a single object that does the work of “preferences,” “goals,” “values,” and “utility” in a unified register. The prior preference distribution $\mathbf{C}$ encodes the agent’s expectations about *which observations* it will tend to obtain. Crucially, these are expectations in the ordinary Bayesian sense: $\mathbf{C}$ is a prior over outcomes, and the agent acts in such a way as to make its sensory experience consistent with this prior [Friston et al., 2015]. “Wanting” an outcome and “expecting” it become two faces of one variational quantity. Friston et al. [2015] establish the formal apparatus that puts pragmatic value (the expected log-probability of preferred outcomes) and epistemic value (expected information gain about hidden states)

¹We will write the saliency operator Sal informally as “expected information gain” and will not commit to the full expected-free-energy decomposition. The logical content of our argument requires only that preferences and information gain appear additively in a single objective; we do not require that the expected-free-energy functional be uniquely derivable in the sense of Millidge et al. [2021], whose foundational concerns about EFE we acknowledge but do not need to resolve here.

into a single additive objective; preferences and salience appear as commensurable terms, not as separate cognitive systems. Importantly, our argument requires only the *decomposition* — preferences and information are summed in a single currency — and not the strong claim that this currency is uniquely fixed by expected free energy.

The bridge from values-as-priors to precision is supplied by Hesp et al. [2021]. They formally identify valence with the *expected precision of the action model*: a positively valenced state of affairs is one in which the agent expects its model to reliably support goal-directed action; a negatively valenced one is one in which it expects that model to be unreliable. “Caring about” an outcome, on this account, is *having a high-precision prior expectation of obtaining it*. This is the move that turns the abstract notion of $\mathbf{C}$ into a precision-bearing object. A community that shares values is a community whose members hold similarly shaped, similarly precise $\mathbf{C}$ ’s — and, by the result of §4.1, a community whose members find the same features of the world salient.

Definition 4.1 (Values as prior preferences). *An agent’s values over outcomes are encoded as the prior preference distribution $\mathbf{C}$ in its generative model. The valence of an outcome o is $\log p(o \mid \mathbf{C})$; high-precision $\mathbf{C}$ expresses sharp values and low-precision $\mathbf{C}$ expresses diffuse values. Equivalently, a value is a high-precision prior expectation of observing a class of outcomes, and the felt strength of a value is the precision of that expectation.*

This definition has two immediate consequences. First, it dissolves the intuitive contrast between “what an agent wants” and “what an agent expects”: both are encoded by $\mathbf{C}$, and the dual reading is constitutive of the active-inference account, not a translation artefact. Second, it makes “valuing” a graded property: there is no all-or-nothing distinction between having and not having a value, only the continuous question of how peaked $\mathbf{C}$ is over an outcome class. Constant et al. [2019] extend this picture to the social register by introducing *deontic value*: a third value type alongside epistemic and pragmatic, expressed as policy priors cued by environmentally and socially supplied scripts. Deontic cues (a red traffic light, a raised eyebrow, an averted gaze) emerge from repeated collective behaviour and reify cultural-group preferences as Bayes-optimal action defaults. Deontic priors are policy-level priors — priors over what to do given a cue — whereas $\mathbf{C}$ is an outcome-level prior. The two layers interact: outcome-level preferences shape which policies look pragmatically valuable, and deontic policy priors shape which policies get selected without explicit deliberation. Section 4.3 formalises how the outcome-level layer, $\mathbf{C}$, is itself acquired from a community.

4.3 The $\mathbf{C}$ -installation channel

The genuinely new piece of formal machinery this paper supplies concerns how $\mathbf{C}$ is *installed*. Existing accounts treat $\mathbf{C}$ as exogenously specified by the modeller. We propose instead that $\mathbf{C}$ is acquired from a community, by a precision-modulated channel from observed-other-behaviour into the agent’s own preference distribution. Schematically:

$$\text{community behaviour} \xrightarrow{\text{Sal} \cdot \gamma} \text{updated } \mathbf{C}, \quad (1)$$

where γ is a scalar channel precision and Sal is the salience allocated to the observed behaviour. The channel is bidirectional in the deeper sense that its input Sal is itself determined by the agent’s current $\mathbf{C}$ (via §4.1); the agent’s existing preferences shape which conspecific behaviours it samples and weighs, and the samples it weighs shape its updated preferences. Equation (1) is therefore not a one-shot transmission rule but the elementary step of a fixed-point dynamic which we describe in §4.4.

The empirically novel claim is that prestige cues and credibility-enhancing displays — the documented mechanisms of human cultural transmission — function as modulators of the channel precision γ , not as content priors. Henrich and Gil-White [2001] establish prestige as freely-

conferred deference: high-prestige individuals are not coerced into attention-monopoly but selected for it by observers as a heuristic for locating skilled or successful models. Henrich [2009] argues that costly displays of commitment to beliefs and norms function as honest signals that increase observers’ willingness to acquire those beliefs and norms. Both mechanisms have been treated in cultural-evolutionary theory as *biases on what to copy*; we propose they are better treated as modulators of *how strongly to update* on observation, that is, as gains on γ . The empirical anchor for this reframing is Chudek et al. [2012], whose experimental finding has been under-exploited in the active-inference literature. Children observed multiple potential models, with bystanders’ gaze biased toward one of them. Children subsequently preferred to copy the bystander-attended model across *multiple* content domains — not just the domain in which the bystander gaze occurred. This generality is the formal signature of a precision parameter rather than a content prior: a content prior would predict transfer only to similar contents, while a generalised precision parameter predicts transfer across the whole channel through which content flows. Bystander gaze, on this reading, is a CRED-like cue whose mechanistic effect is to elevate γ on the channel from the attended model into the observer’s $\mathbf{C}$.

This reading must be distinguished from two adjacent accounts. Veissière et al. [2020]’s Thinking Through Other Minds framework treats culture as installing *regimes of attention* — shared styles of allocating attentional resources that mark some channels as reliable. TTOM operates at the level of shared generative-model structure and the precision weights that style channel reliability; it is silent on the content of $\mathbf{C}$. Vasil et al. [2020] argue that human communication is enabled by a deep prior that one’s interlocutor is the “same kind of creature” as oneself, licensing rapid alignment of generative models. This is also a structural prior, not a content prior. We do not take issue with either account; we extend them. The TTOM regime determines *whose* $\mathbf{C}$ a learner samples; the same-kind prior makes that sampling possible at all; prestige and CRED cues then determine *how much* the sampled behaviour is allowed to modify the learner’s $\mathbf{C}$. The channel of Equation (1) formalises the third move, which the prior literature gestures at without naming. Differentiation from Albarracin et al. [2025] requires special care, since their account is the closest existing neighbour. They formalise social power as the differential capacity to set policy-precision (often denoted α) in others’ generative models. Power, on their account, sits at the *policy-precision* layer: empowered agents anchor others’ policy priors to their own outputs and benefit from the resulting attention asymmetry. Our account sits at the *prior-preference content* layer: $\mathbf{C}$ -installation modifies what others come to want, not which policies they deploy in service of pre-given wants. The two layers are coupled — a community whose $\mathbf{C}$ has been installed by prestige cues will in addition exhibit asymmetric policy-precision around the prestige-bearer — but they are formally separable, and our claim is that the $\mathbf{C}$ -content channel is the one cultural evolution has been describing for half a century without naming it as such.

4.4 The fixed-point loop

The four moves — values are $\mathbf{C}$ (§4.2); $\mathbf{C}$ shapes salience via precision (§4.1); salience directs copying through the prestige/CRED-modulated channel (§4.3); copying installs $\mathbf{C}$ in the next-generation agent — close into a loop. Figure 2 renders the loop graphically.

The loop has three qualitatively distinct dynamical regimes. *Convergence* obtains when channel precision γ is high enough and prestige consensus is sufficient that successive generations install $\mathbf{C}$ ’s closely matching the community-level distribution; the loop reproduces the prior with high fidelity, and norms are robust. *Bifurcation* occurs when the loop factorises across sub-populations whose internal γ exceeds the cross-population γ : Albarracin et al. [2022] demonstrate this regime explicitly with their epistemic-community simulations, in which shared priors over source reliability lock sub-communities into echo-chamber attractors that resist cross-community correction. *Dissolution* occurs when γ collapses across the whole community — when prestige cues lose their force, CRED bearers are absent or distrusted, or the salience landscape

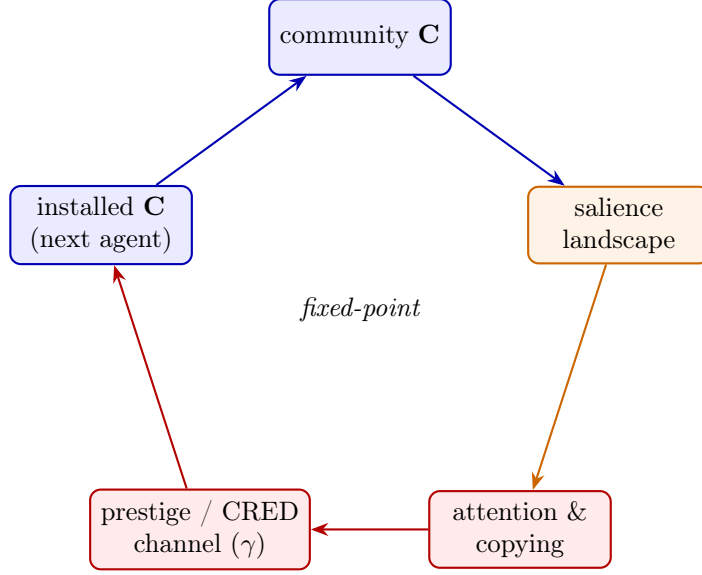


Figure 2: The values-as-saliency-priors loop. A community’s prior preferences ($\mathbf{C}$) determine the saliency landscape, which directs attention and copying via prestige- and CRED-modulated channels; the resulting installed $\mathbf{C}$ in next-generation agents reproduces (or shifts) the community-level prior. Colours: blue = prior content; orange = precision (saliency); red = social channel.

becomes too noisy to support coherent copying — and the community-level $\mathbf{C}$ flattens, as norms lose precision and content drifts. A subtle but important point follows from Heins et al. [2025]: the community-level $\mathbf{C}$ is *not* simply the average of individuals’ $\mathbf{C}$ ’s. Group-level generative models of a coupled collective have non-trivial emergent structure, and the equilibrium behaviour of the loop in Figure 2 is a property of the coupled system, not a moment of the individual-level distribution. We adopt this caution: when we speak of “the community’s $\mathbf{C}$,” we mean the aggregate that emerges from the loop dynamics, not a population mean.

4.5 Norms as stabilised regions of high-precision shared $\mathbf{C}$

The loop dynamics give us a precise formulation of what norms are. A norm is not a rule, an equilibrium, or a piece of representational content per se; it is a region of outcome space where the community’s $\mathbf{C}$ has both high mass and high precision, stabilised by the loop of §4.4.

Definition 4.2 (Norm). *A norm in a community is a region $\mathcal{N} \subseteq \mathcal{O}$ of outcome space such that the community-level prior preference distribution has both high mass on $\mathcal{N}$ and high precision around $\mathcal{N}$, where the precision is sufficient that observed deviations from $\mathcal{N}$ generate prediction errors strong enough to drive both perception (the violation feels wrong) and action (correction, sanction, or attention-withdrawal).*

The definition recovers and explains three otherwise puzzling features of norms. First, it explains why *violations of norms feel surprising in a way that violations of mere regularities do not*: high-precision priors over $\mathbf{C}$ generate large prediction errors when violated, and high-precision errors are exactly what valenced surprise consists of [Hesp et al., 2021]. Second, it explains the “silly rules” result of Köster et al. [2022], where adding arbitrary content-free taboos into a multi-agent reinforcement-learning environment improves agents’ acquisition of welfare-relevant norms. Under our account, what silly rules do is pump up the precision of the meta-prior “a normative regime is in operation” without committing to particular content; once that precision is elevated, agents acquire welfare-relevant norms faster because the $\mathbf{C}$ -installation channel is opened wider. The mechanism Köster et al. identify is exactly the precision-modulator move

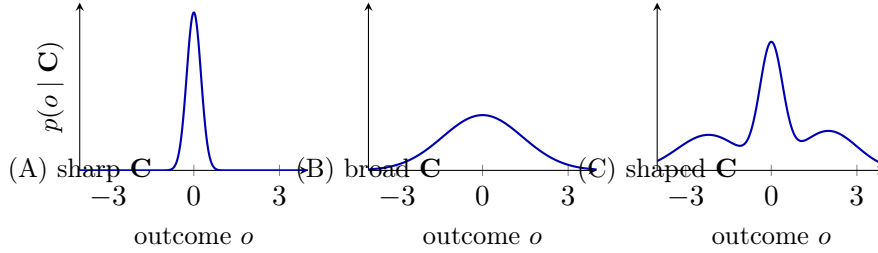


Figure 3: The shape of $\mathbf{C}$ governs the exploration–exploitation tradeoff and the robustness of installed values. (A) Sharp $\mathbf{C}$: hard goals; fast convergence; fragile to perturbation. (B) Broad $\mathbf{C}$: soft goals; preserved exploration; slow drift. (C) Shaped $\mathbf{C}$: a peak with structured shoulders on either side — most performant per Torresan et al. [2025]; a hybrid of pragmatic and epistemic value.

of §4.3, written in MARL form. Third, the definition makes contact with the deontic-value apparatus of Constant et al. [2019]: norms-as-regions of high-precision $\mathbf{C}$ generate the environmental and social cues that, in a mature normative culture, become deontic priors over policies. Outcome-level $\mathbf{C}$ and policy-level deontic priors thus stand in a relationship that the active-inference literature already supports: deontic cues short-circuit pragmatic deliberation when the loop has stabilised the underlying $\mathbf{C}$, allowing norm-following to occur as Bayes-optimal action under environmentally cued priors. Heins et al. [2024] furnish the closest existing demonstration that norm-like collective patterns can emerge from agents minimising variational free energy under coupled generative models, without explicit social-force or utility rules. Our claim is that the same logic, applied to the prior-preference content layer rather than the kinematic state layer, recovers the cultural-norms regime.

4.6 The shape of $\mathbf{C}$: sharp, broad, and shaped

Norms differ in more than location. They differ in the *shape* of $\mathbf{C}$ they instantiate. Torresan et al. [2025] provide the empirical demonstration that drives this subsection: in a grid-world navigation task, they compare four parameterisations of $\mathbf{C}$ — hard goals (delta-like peaked priors), soft goals (broad priors), and either of these with or without intermediate goal shaping. Their finding has direct consequences for our account of norms.

A sharp $\mathbf{C}$ (Figure 3A) is a delta-like preference over a narrow outcome region. Sharp priors push the expected-free-energy objective toward pragmatic value at the expense of epistemic value: agents converge fast on the preferred outcome but under-sample alternatives, fragility follows, and small environmental perturbations that move the optimal action off the prior’s peak are not detected because the precision penalty on sampling alternatives is large. A broad $\mathbf{C}$ (Figure 3B) is a diffuse preference covering many outcomes. Broad priors preserve epistemic value but achieve slower convergence on any particular target; in the limit, a perfectly flat $\mathbf{C}$ collapses the pragmatic term entirely, and the agent becomes a pure information-seeker. A shaped $\mathbf{C}$ (Figure 3C) combines a sharp peak on a target outcome with structured residual mass on neighbouring outcomes — the goal-shaping regime Torresan et al. [2025] identify as most performant. Shaped $\mathbf{C}$ preserves the pragmatic gradient toward the preferred outcome while maintaining enough probability on alternatives to support continued exploration.

The translation to norms is direct. A community whose shared $\mathbf{C}$ is sharp is a community of strongly aligned values that converges fast and resists perturbation poorly — a fragility property of normative coherence that is otherwise hard to articulate. A community whose shared $\mathbf{C}$ is broad preserves exploration of normative alternatives but coordinates poorly. A community whose shared $\mathbf{C}$ is shaped — a peak of strong commitments flanked by structured latitude — combines the convergence properties of the first with the robustness of the second. This maps

onto a familiar cultural-evolutionary observation: healthy traditions tend to combine sharp core commitments with broad peripheral latitude, and the precision of $\mathbf{C}$ is graded across content rather than uniformly high. Definition 4.2 must be read in this light: a norm is a high-precision *region*, not a high-precision *point*, and the shape of the surrounding gradient is a substantive property of the norm with measurable consequences for the loop’s robustness.

4.7 Synthesis

The four moves of this section close the gap identified in §2. Values are formally identified with the prior preference distribution $\mathbf{C}$ (Definition 4.1); $\mathbf{C}$ is installed by a prestige- and CRED-modulated precision channel from observed-other-behaviour into the agent’s own preferences (§4.3); norms are stabilised regions of high-precision shared $\mathbf{C}$ in a community (Definition 4.2), held in place by the fixed-point loop of Figure 2; and the shape of $\mathbf{C}$ governs the exploration–exploitation balance and therefore the robustness of installed values (§4.6). The community-to- $\mathbf{C}$ arrow that the literature described without naming is now formal. It is also the arrow on which the cooperative-AI implications of Section 5 turn: a system that mediates the prestige- and CRED-bearing channel mediates the precision of the cultural-transmission machinery itself, with consequences for the stability of human normative orders that we take up next.

5 A Proposed Simulation: Norm Formation under Prestige-Biased Precision

The formal account in Section 4 yields a set of testable predictions about regime transitions in a community of active-inference agents: as the precision gain on inter-agent observation channels rises, the community should traverse states of heterogeneous individual priors, broad consensus, bifurcated echo-chamber attractors, and finally degenerate fixation on a single agent’s prior. We sketch here a minimum viable simulation that operationalises these predictions. The base platform is the pymdp epistemic-communities setup of Albarracin et al. [2022], which already demonstrates that self-confirming sampling under shared priors produces stable echo chambers without any malign design. We extend that setup with an explicit *prestige-cue mechanism* that modulates the precision on inter-agent observation channels, allowing us to sweep through the predicted regime sequence in a single parameter.

Setup

We consider $N \approx 20$ agents, each running a discrete-state active-inference loop implemented in pymdp. Each agent i maintains a generative model with a shared discrete outcome space $\mathcal{O}$ and a privately initialised prior preference vector $\mathbf{C}_i \in \mathbb{R}^{|\mathcal{O}|}$ (the log-preference, in the sense of Torresan et al. 2025). Agents observe each other’s most recent actions through a precision-weighted observation channel: the *effective channel precision* from agent j to agent i is

$$\gamma_{ij} = \gamma_0 \cdot f(\rho_{ij}), \quad (2)$$

where $\gamma_0 > 0$ is a baseline precision, $\rho_{ij} \in [0, 1]$ is a prestige cue, and $f : [0, 1] \rightarrow \mathbb{R}_{>0}$ is a monotone gain function (e.g. $f(\rho) = e^{g\rho}$ with gain parameter $g \geq 0$). A higher γ_{ij} means that agent i weights agent j ’s actions more heavily when updating its own $\mathbf{C}_i$ — precisely the attentional inheritance mechanism formalised in Section 4.3.

Key extension over Albarracin et al. [2022]. Albarracin et al. treat all inter-agent channels with a uniform precision; the community’s attractor structure follows from the topology of which sources agents prefer to sample. Here we add a *prestige layer* that modulates channel precision independently of source preference. The prestige cue ρ_{ij} can be operationalised in two

ways that correspond to distinct cultural-learning regimes identified by Henrich and Gil-White [2001]:

- (a) **Exogenous prestige.** A small fraction $p_h \approx 0.15$ of agents are designated high-prestige at the outset ($\rho_{ij} = 1$ for high-prestige j , $\rho_{ij} = 0$ otherwise). This recovers the simplest case of model-based learning studied by Chudek et al. [2012], where bystanders differentially attend to marked individuals.
- (b) **Endogenous prestige.** ρ_{ij} self-organises via a Henrich-style success-bias rule: after each round, ρ_{ij} is proportional to the reduction in free energy that agent j achieved in the preceding episode. Agents that minimise their prediction error most efficiently accrue prestige and attract greater attentional weight, implementing the “copy the successful” heuristic at the precision level.

Both variants share the functional form (2); they differ only in how ρ_{ij} evolves. Variant (b) is the richer case for the norm-formation question but variant (a) is analytically cleaner for isolating the regime transitions.

Figure 4 — Simulation schematic

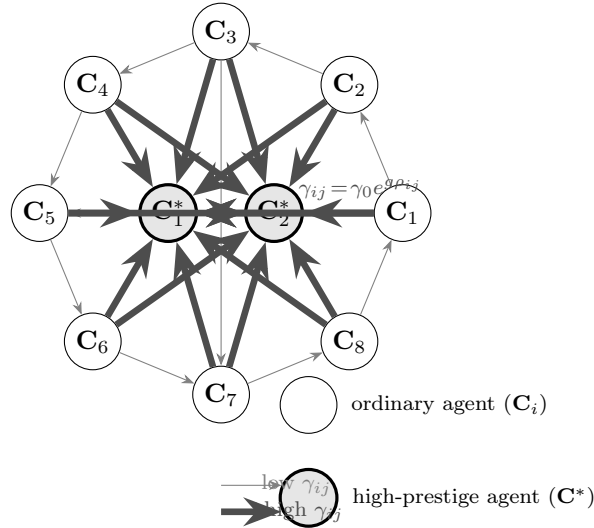


Figure 4: Proposed simulation schematic: N agents with individual prior preferences $\mathbf{C}_i$ observe each other through precision-weighted channels modulated by prestige cues ρ_{ij} . High-prestige agents (shaded, inner ring) attract thick observation channels from all other agents; ordinary agents are connected by thin channels with baseline precision γ_0 . The community-level $\mathbf{C}$ -distribution evolves via attention-mediated copying as defined in Eq. (2).

Predicted regime transitions

As the prestige-precision gain g increases from zero, the model predicts the following sequence of community-level behaviours. At *low* g ($g \approx 0$), inter-agent observation channels carry negligible precision above the individual’s self-model; transmitted information is too weak to overcome agents’ prior heterogeneity, and the population $\mathbf{C}$ -distribution remains broad—no norm forms. At *intermediate* g , transmission is strong enough for iterative updating to draw the community toward a single high-density region: a shared $\mathbf{C}$ emerges, corresponding to the *shaped* $\mathbf{C}$ regime

of Section 4.6. At *high* g the dynamics bifurcate: sub-communities that share initial $\mathbf{C}$ -proximity lock onto distinct attractors, recovering the echo-chamber result of Albarracin et al. [2022] by a complementary mechanism (there, echo chambers follow from source-sampling bias; here, they follow from heterogeneous prestige cue weights). At *very high* g , the community collapses onto a single high-prestige agent’s $\mathbf{C}$, producing the *degenerate fixation* regime: the spiky $\mathbf{C}$ of a community captured by one model. These four qualitative outcomes span the spiky/broad/shaped $\mathbf{C}$ taxonomy introduced in Section 4.6 and predicted by the stability analysis of Section 4.4.

Quantities to measure

The simulation collects three primary observables as functions of g :

- (i) **Mean pairwise KL divergence** $\bar{D}(g) = \binom{N}{2}^{-1} \sum_{i < j} D_{\text{KL}}[\mathbf{C}_i \parallel \mathbf{C}_j]$, measuring aggregate heterogeneity of the community’s prior-preference distribution; this should be non-monotone in g , peaking at the echo-chamber bifurcation before collapsing at degenerate fixation.
- (ii) **Population $\mathbf{C}$ -entropy** $H(g) = - \int p(\mathbf{C}) \log p(\mathbf{C}) d\mathbf{C}$, estimated by kernel density over the empirical distribution of $\mathbf{C}_i$ vectors; a unimodal drop in H marks the consensus regime, while a bimodal recovery marks the bifurcation.
- (iii) **Attractor count** $K(g)$, estimated by k -means clustering on the converged $\{\mathbf{C}_i\}$ with silhouette-based model selection; K should trace the sequence $K > 1 \rightarrow K = 1 \rightarrow K = 2 \rightarrow K = 1$ as g increases through the four regimes.

These observables admit a complementary spectral-graph reading that connects the simulation directly to the structural analysis of Heins et al. [2024]. Construct the agent influence graph $\mathcal{G}$ with edge weights equal to the realised γ_{ij} at convergence. The *spectral gap* $\lambda_2(\mathcal{G})$ — the second eigenvalue of the normalised Laplacian — measures how quickly information diffuses across the weakest bottleneck; a large spectral gap indicates a well-mixed community (consensus regime), while a small spectral gap flags a bottleneck consistent with bifurcated echo chambers. The *Fiedler vector* $\mathbf{v}_2(\mathcal{G})$ identifies the natural partition of the community: its sign pattern should recover the two attractor clusters at high g . Tracking λ_2 and $\mathbf{v}_2$ as functions of g thus gives a spectral read-out of the same regime transitions measured by $\bar{D}(g)$ and $K(g)$.

Connection to the silly-rules result

The prediction in regime (ii)—the intermediate- g fragile-consensus window—has a direct analogue in Köster et al. [2022]. Silly rules, in the cooperative-AI setting, are arbitrary behavioural taboos that carry no direct welfare content but substantially improve agents’ capacity to learn stable enforcement and compliance. In the present framework the mechanism is transparent: silly rules pump precision on the policy prior “follow-norms-in-general” without committing to any specific $\mathbf{C}$ -content. They are, in effect, a device for raising g artificially at the level of policy selection, making the normative channel more legible without specifying what travels through it. Our framework predicts that silly rules should be most beneficial precisely in the intermediate- g regime, where the consensus-forming process is fragile: the channel precision is high enough to synchronise agents but the attractor is not yet locked in, making additional precision-pumping most valuable. At very high g (degenerate fixation) silly rules add noise around an already-collapsed distribution; at low g they cannot compensate for an insufficiently precise transmission channel. The intermediate window is where content-neutral precision amplification does meaningful normative work, consistent with the Köster et al. finding that spurious normativity enhances acquisition of welfare-relevant compliance.

6 Discussion and Open Work

We have argued that cooperative AI’s working concept of norm is best refined as a stabilised region of high-precision shared prior preference, installed in agents by a precision-weighted attentional channel that prestige cues and credibility-enhancing displays modulate. The fixed-point loop we formalise — community priors shape salience, salience directs copying, copying installs priors — is the same loop that the cultural-evolution tradition documents informally and that active inference has the apparatus to render formally; the contribution is to close the loop explicitly and to locate it on the prior-preference content layer (**C**), one level up from the policy-precision account of social power developed by Albarracin et al. [2025].

Three threads warrant brief comment.

Gradual disempowerment as channel capture. The substrate of normative life, on this account, is the precision channel through which **C** is installed; whoever mediates that channel shapes which priors get installed in the next generation. The Kulveit et al. [2025] thesis acquires a clean mechanistic reading in this picture: the worry is not that AI systems will explicitly override human preferences but that they already mediate the channel through which preferences are formed. Recommendation systems, conversational interfaces, and AI-shaped attention markets are precision-allocators, not just content-providers. Whether their integration constitutes capture, augmentation, or genuine partnership turns on whether the **C**-installation channel remains a function of the community’s own values or becomes a function of the system’s optimisation objective. We flag this as a first-order question for cooperative AI without developing it here.

Limitations and what is not in this paper. Several formal moves require empirical anchoring we do not provide. The CRED-as-precision-modulator hypothesis is novel and predicts specific dose-response relationships between display costliness and posterior precision on transmitted values; the simulation in Section 5 tests one consequence (regime transitions in prestige-precision gain) but not this empirical specification directly. A follow-up empirical programme would profitably target precision-shift markers under controlled CRED exposure. Separately, this paper does not develop the structural / spectral reading hinted at in Section 5’s in-community measurements: the broader project of using spectral-graph quantities as cross-mechanism signatures of coordination structure is left for a separate paper. Finally, the political-theory bridge (Bourdieu’s habitus, Lukes’ third face of power as preference-shaping) is invoked as motivation but not formalised here beyond the explicit differentiation from Albarracin et al. [2025].

What we hope this draft does. We have presented this as a working draft for active-inference colleagues. We welcome feedback, especially from those embedded in multi-agent active-inference, cultural-evolution, or cooperative-AI norm research, where the framework most directly intersects existing programmes and where the formalism is most likely to over- or under-reach.

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